CAMPUS AI

Intelligent Academic Management System

1.**GitHub URL: https://github.com/vijaybabu2204/CampusAI.git**

2. Project Overview

2.1 Problem Statement

In most educational institutions, schedule management is largely manual and fragmented. Sudden changes — room modifications, session cancellations, venue shifts, and special events — are often communicated through notice boards, WhatsApp groups, or word-of-mouth, leading to:

- Students missing important classes due to room changes
- Delays in receiving emergency schedule modifications
- No centralized source of truth for academic timetables
- Admins spending significant time on repetitive communication tasks
- Students unaware of upcoming placement drives and opportunities

2.2 Proposed Solution

Campus AI provides a centralized, AI-driven platform that solves these challenges through:

- An Admin Agent that processes natural language instructions and automatically propagates schedule changes
- A Student Agent that answers academic queries in real time using the most up-to-date data
- Automated Power Automate workflows that handle data operations in the background
- A hybrid Dataverse database storing both regular and emergency schedule data

2.3 Project Objectives

1. Eliminate manual schedule communication through AI-powered automation
2. Provide students with 24/7 access to accurate schedule and placement information
3. Enable administrators to update schedules using simple natural language commands
4. Maintain a single centralized data store accessible by all AI agents and users
5. Improve the overall student experience through real-time, intelligent notifications

2.4 Scope

Campus AI covers the following functional areas:

- Emergency schedule update management (cancellations, room changes, venue updates)
- Student schedule query handling (class locations, trainer info, session timings)
- Placement drive information dissemination
- Automated backend data operations (insert, retrieve, override, delete)
- Dashboard and analytics via Power BI

3. System Architecture

3.1 Architecture Overview

Campus AI follows a three-tier architecture pattern built entirely on Microsoft's cloud-based Power Platform ecosystem. The three tiers consist of the Presentation Layer (Power Apps + Copilot Studio agents), the Logic Layer (Power Automate workflows), and the Data Layer (Microsoft Dataverse).

3.2 Architecture Layers

Presentation Layer

Component	Technology	Purpose
Admin Interface	Microsoft Copilot Studio	Admin Agent for processing natural language schedule commands
Student Interface	Microsoft Copilot Studio	Student Agent for answering academic and placement queries
Management Dashboard	Microsoft Power Apps	Visual interface for admins to manage schedule data
Analytics Dashboard	Microsoft Power BI	Reports and visual insights on schedule data

Logic Layer

All backend operations are handled by Power Automate cloud flows that are triggered by agent actions or scheduled timers:

- Add Emergency Update Flow — triggered when the Admin Agent receives a schedule change command
- Retrieve Schedule Flow — triggered when the Student Agent requests timetable data
- Override Logic Flow — applies emergency updates over the regular weekly schedule
- Cleanup Flow — scheduled trigger to automatically delete expired emergency update records

Data Layer

Microsoft Dataverse serves as the central cloud database with two primary tables:

- Weekly Schedule Table — stores the regular recurring academic timetable
- Emergency Updates Table — stores temporary modifications and overrides

3.3 Data Flow

6. Admin issues a natural language command to the Admin Agent
7. Admin Agent processes the instruction using NLP and extracts key parameters
8. Admin Agent triggers the Power Automate flow to write data to Dataverse
9. Student queries the Student Agent conversationally
10. Student Agent triggers the retrieval flow and checks both Dataverse tables
11. If an emergency update exists, it overrides the regular timetable
12. Student receives the most accurate and up-to-date response

4. Technology Stack

Campus AI is built entirely on Microsoft's Power Platform ecosystem, enabling a no-code/low-code development approach while leveraging enterprise-grade AI and cloud capabilities.

Technology	Component	Role in Campus AI
Microsoft Power Apps	Frontend UI	Administrative canvas app for managing schedules and viewing data
Microsoft Copilot Studio	AI Agents	Building and deploying the Admin Agent and Student Agent
Microsoft Power Automate	Workflow Engine	Automating data operations, overrides, and cleanup flows
Microsoft Dataverse	Cloud Database	Central storage for Weekly Schedule and Emergency Updates tables
Microsoft Power BI	Analytics	Visual dashboards and reports for schedule and attendance data

4.1 Why Microsoft Power Platform?

- Rapid development with minimal coding using visual designers
- Native integration between all platform components (no API development needed)
- Enterprise-grade security and compliance with Azure Active Directory
- Built-in AI capabilities through Copilot Studio's NLP engine
- Scalable cloud infrastructure managed by Microsoft Azure
- Familiar ecosystem for institutions already using Microsoft 365

5. AI Agents

Campus AI consists of two purpose-built AI agents created in Microsoft Copilot Studio. Each agent handles a distinct role within the system and communicates with the backend through Power Automate workflows.

5.1 Admin Agent

Overview

The Admin Agent is the administrative backbone of Campus AI. It enables college administrators to issue schedule updates using plain English, without needing to interact with any forms or databases directly. The agent accepts natural language instructions, extracts structured data using NLP, and automatically stores it in the Dataverse Emergency Updates table via Power Automate.

Sample Commands

- "Move Flutter class to Bay-3"
- "Cancel today's sessions"
- "No classes for DRIVE READY batch tomorrow"
- "Change Python room to Lab-2 from Monday to Wednesday"
- "Postpone all afternoon sessions for Batch A today"

NLP Parameter Extraction

When an admin issues a command, the Admin Agent extracts the following structured parameters using Natural Language Processing:

Parameter	Description	Example Value
Batch Name	The student batch affected by the update	DRIVE READY, Batch-A, Full Stack
Technology	The subject or technology domain	Flutter, Python, AWS, React
Room / Venue	The physical or virtual location	Bay-3, Lab-2, Room 104
Action Type	The type of schedule change	Cancellation, Room Change, Postponement
Start Date	Effective start date of the update	Today, 2025-06-10
End Date	Expiry date of the emergency update	Tomorrow, 2025-06-12

Admin Agent Flow

13. Admin types or speaks a natural language command
14. Copilot Studio processes input and runs NLP entity extraction
15. Extracted parameters are validated and confirmed
16. Power Automate flow is triggered with the structured payload
17. Data is written to the Emergency Updates table in Dataverse
18. Admin receives confirmation of the update

5.2 Student Agent

Overview

The Student Agent is the primary student-facing interface of Campus AI. Students interact with it through a conversational interface to get instant answers about their academic schedules, class locations, trainer details, and upcoming placement drives. The agent retrieves data from both Dataverse tables and applies override logic to ensure students always receive the most current information.

Sample Student Queries

- "Where is today's Python class?"
- "Who is the trainer for Flutter?"
- "Any updates today?"
- "When is the AWS session scheduled?"
- "Any placement drives this week?"
- "What time does the React class start?"

Override Logic

The Student Agent uses a smart override mechanism to deliver accurate information:

19. Student submits a query to the Student Agent
20. Agent triggers the retrieve schedule Power Automate flow
21. Flow checks the Emergency Updates table for any active overrides
22. If an emergency update exists — it takes priority over the regular schedule
23. If no emergency update — the agent returns data from the Weekly Schedule table
24. Student receives a clear, accurate response with the latest information

Query Categories Supported

Query Type	Example Query	Data Source
Room / Location	Where is today's Python class?	Emergency Updates → Weekly Schedule
Trainer Info	Who is the trainer for Flutter?	Weekly Schedule Table
Schedule Timing	When is the AWS session?	Weekly Schedule Table
Emergency Updates	Any updates today?	Emergency Updates Table
Placement Drives	Any drives this week?	Placement Drives Table
Cancellations	Are classes cancelled today?	Emergency Updates Table

6. Database Design

Campus AI uses a hybrid database architecture in Microsoft Dataverse. Rather than storing all data in a single table, the system separates regular schedule data from emergency modifications to avoid redundancy, improve query performance, and simplify the override logic.

6.1 Weekly Schedule Table

This table stores the permanent, recurring academic timetable for all batches. It is the base data source for the Student Agent.

Field Name	Data Type	Description
Batch Name	Text	Name of the student batch (e.g., Full Stack, DRIVE READY)
Technology	Text	Subject or technology module (e.g., Python, Flutter, AWS)
Trainer Name	Text	Name of the assigned trainer or faculty
Room / Venue	Text	Default classroom or venue for the session
Day of Week	Text	Day(s) on which the session is scheduled
Start Time	Date/Time	Time the session begins
End Time	Date/Time	Time the session ends

6.2 Emergency Updates Table

This table stores all temporary schedule modifications. When an admin issues a change command, the data lands here and is used to override the Weekly Schedule when queried.

Field Name	Data Type	Description
Update ID	Auto Number	Unique identifier for the emergency update record
Batch Name	Text	Batch affected by the emergency update
Technology	Text	Subject or technology module impacted
Action Type	Choice	Type of update: Cancellation, Room Change, Venue Shift
New Room / Venue	Text	Updated location (if applicable)
Start Date	Date Only	Date from which the emergency update is effective
End Date	Date Only	Date on which the emergency update expires
Created On	Date/Time	Timestamp when the update was created

6.3 Database Design Rationale

- Separation of concerns — regular data and emergency data are independent
- Avoids redundant data — only changed fields are stored in Emergency Updates
- Automatic cleanup — expired records are deleted by scheduled Power Automate flows
- Scalable — new table types can be added without restructuring existing schema
- Query efficiency — agents query Emergency Updates first, reducing unnecessary lookups

7. Power Automate Workflows

Power Automate cloud flows form the backend engine of Campus AI. They are triggered by AI agent actions or scheduled timers and handle all data operations between the agents and Dataverse.

Flow Name	Trigger	Action	Used By
Add Emergency Update	Admin Agent command	Insert record into Emergency Updates table	Admin Agent
Retrieve Student Schedule	Student Agent query	Query Weekly Schedule & Emergency Updates, apply override	Student Agent
Apply Override Logic	Called by Retrieve Flow	Compare dates and return active emergency data	Student Agent
Delete Expired Updates	Scheduled (daily timer)	Remove past-date records from Emergency Updates table	System (automated)
Placement Drive Query	Student Agent query	Return active placement drive records	Student Agent

7.1 Add Emergency Update Flow

Triggered when the Admin Agent submits a structured update payload. The flow validates the input, maps it to the Emergency Updates table schema, and performs an insert operation in Dataverse.

7.2 Retrieve Schedule Flow

Triggered by the Student Agent when a student submits a query. The flow:

25. Checks the Emergency Updates table for active records matching the query criteria
26. If found — returns the emergency data (room change, cancellation, etc.)
27. If not found — returns the matching record from the Weekly Schedule table
28. Formats the response for the Student Agent to deliver conversationally

7.3 Cleanup Flow (Scheduled)

Runs daily on an automated schedule. Scans the Emergency Updates table for records where the End Date is before today and permanently deletes them. This keeps the table lean and prevents stale data from affecting student queries.

8. Key Features

8.1 Real-Time Schedule Communication

Administrators can update schedules in seconds using plain English. Changes propagate instantly to the Student Agent, ensuring students get live information without any delay.

8.2 AI-Powered Conversational Interface

Both the Admin Agent and Student Agent are powered by Microsoft Copilot Studio's NLP engine. Users interact naturally — no training, forms, or technical knowledge required.

8.3 Intelligent Override Logic

When emergency updates exist, the system automatically prioritizes them over the regular timetable. Students never need to cross-reference multiple sources.

8.4 Placement Drive Integration

Campus AI includes a dedicated placement drive module. Students can ask the Student Agent about upcoming company visits, drive schedules, eligibility criteria, and deadlines — all from the same conversational interface.

8.5 Automated Data Cleanup

Expired emergency updates are automatically removed daily by a scheduled Power Automate flow. This ensures data accuracy without any manual maintenance.

8.6 Analytics and Reporting

Power BI dashboards provide administrators with visual reports on schedule changes, cancellation frequency, batch-wise updates, and placement drive participation — enabling data-driven decision-making.

9. Non-Functional Requirements

Category	Requirement	Details
Performance	Response Time	Student Agent responses delivered in under 3 seconds
Availability	Uptime	System available 24/7 via Microsoft Azure cloud infrastructure
Scalability	User Load	Supports concurrent usage by all students and admins simultaneously
Security	Authentication	Secured via Azure Active Directory and Microsoft 365 credentials
Data Integrity	Accuracy	Override logic ensures students always receive the most current data
Maintainability	No-Code Updates	Flows and agents can be updated without developer involvement
Compliance	Data Governance	Dataverse enforces role-based access control (RBAC)

10. Future Enhancements

The following enhancements are planned for future versions of Campus AI:

- Push notifications via Microsoft Teams and email for schedule changes
- Multi-language support for regional student populations
- Integration with biometric attendance systems

- Voice-based interaction support for the Student Agent
- Mobile application via Power Apps mobile for students
- AI-driven predictive analytics for schedule conflict detection
- Integration with LMS platforms such as Moodle or Blackboard
- Student feedback collection through post-session surveys

11. Conclusion

Campus AI demonstrates how modern AI, cloud automation, and centralized data management can be combined to solve a real and pressing problem in educational institutions. By replacing manual, fragmented communication with an intelligent, automated platform, Campus AI delivers measurable improvements in efficiency, accuracy, and the overall student experience.

The system's architecture — powered entirely by Microsoft's Power Platform — ensures that it is scalable, secure, and maintainable without heavy development overhead. The two AI agents at its core bring natural language interaction to both administrators and students, making the platform accessible and easy to adopt.

Campus AI sets a strong foundation for the future of smart campus ecosystems, where AI-driven automation seamlessly connects students, faculty, and administrators in real time.

Campus AI | Intelligent Academic Management System
Built on Microsoft Power Platform